Programming Fundamentals

Lab 14	
Topic	Double Pointer and Dynamic Memory Allocation
Objective	Learning objectives of this lab are to get hands on experience on utilizing the concepts of dealing with the memory addresses and manage data given in tabular form. After successfully accomplishing this lab, students will be able to understand the • Need of double pointer • Difference between single and double pointers • Dynamic Memory allocation involving double pointer • Creating 2D array dynamically using double pointer • Explaining benefits of creating 2D arrays dynamically • Problem solving involving dynamically allocated 2D arrays and functions

Task 1

Write a program in C++ that reads two matrices from a file and calculates the sum of two matrices.

Example

Data.txt

```
2 3
1 2 3
4 5 6
2 3
9 7 4
1 2 8
```

$$A+B=\begin{bmatrix}1 & 2 & 3\\4 & 5 & 6\end{bmatrix}+\begin{bmatrix}9 & 7 & 4\\1 & 2 & 8\end{bmatrix}=\begin{bmatrix}10 & 9 & 7\\5 & 7 & 14\end{bmatrix}$$

Task 2

Write a program in C++ that reads a matrix from a file, and displays the diagonal of a matrix.

Example

Data.txt

```
3 3
1 2 3
4 5 6
7 8 9
```

$$A=\begin{bmatrix}1 & 2 & 3\\ 4 & 5 & 6\\ 7 & 8 & 9\end{bmatrix}$$

Output on console 1 5 9

Task 3

Write a program in C++, that reads a matrix from a file, and displays the data column wise.

Example

Data.txt

```
2 3

1 2 3

4 5 6
```

$$A=\begin{bmatrix}1 & 2 & 3\\ 4 & 5 & 6\end{bmatrix}$$

Output on console: $\begin{matrix}1 & 4\\ 2 & 5\\ 3 & 6\end{matrix}$

Task 4

Write a program in C++ that reads the data from a file and sorts each row of a matrix.

Example

Data.txt

```
3 4

12 55 7 66

56 4 89 14

33 41 32 53
```

Original Matrix: $A=\begin{matrix}12 & 55 & 7 & 66\\ 56 & 4 & 89 & 14\\ 33 & 41 & 32 & 53\end{matrix}$

Updated Matrix: $A=\begin{matrix}7 & 12 & 55 & 66\\ 4 & 14 & 56 & 89\\ 32 & 33 & 41 & 53\end{matrix}$

Task 5

Write a program that reads a data from a file and calculates the sum of data row-wise and column-wise.

Example

Data.txt

```
2 3

1 2 3

4 5 6
```

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$$

Sum row-wise: 6, 15

Sum column-wise: 5, 7, 9

Task 6

Write a program in C++ that reads the data from a file and rotates the matrix clockwise by 90°.

Example

Data.txt

```
3

1 2 3 -99

4 5 6 -99

7 8 9 -99
```

Original Matrix: $A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$

Updated Matrix: $A = \begin{bmatrix} 7 & 4 & 1 \\ 8 & 5 & 2 \\ 9 & 6 & 3 \end{bmatrix}$

Task 7

Write a program that reads two matrices from a file and calculates the dot product and cross product of two matrices.

Example

Data.txt

```
2
1 2 3 -99
4 5 6 -99
2
9 7 4 -99
1 2 8 -99
```

$$A.B=\begin{bmatrix}1 & 2 & 3\\4 & 5 & 6\end{bmatrix}.\begin{bmatrix}9 & 7 & 4\\1 & 2 & 8\end{bmatrix}=\begin{bmatrix}9 & 14 & 12\\4 & 10 & 48\end{bmatrix}$$

Example (For cross multiplication, take transpose of second matrix)

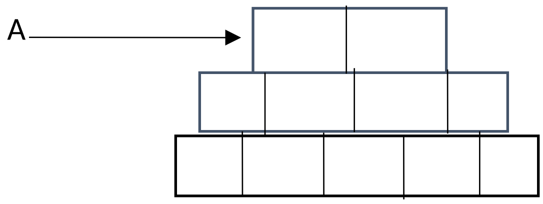
Jagged Edges	
Objective	• 2D Array with Jagged edges o Discuss need of jagged array. o Dynamic Memory allocation for jagged edges involving double pointer. o Creating 2D array dynamically for jagged edges using double pointer. o Explaining benefits of creating 2D arrays for jagged edges dynamically. o Discuss why jagged array is not possible as static memory. o Problem solving involving jagged edges, dynamically allocated 2D arrays and functions.

$$A * B=\begin{bmatrix}1 & 2 & 3\\4 & 5 & 6\end{bmatrix}+\begin{bmatrix}9 & 1\\7 & 2\\8 & 4\end{bmatrix}=\begin{bmatrix}47 & 17\\119 & 38\end{bmatrix}$$

Lab Tasks

Lab Task-1

A Cinema Hall consist of 10 rows. Each row is assigned an Alphabet. There are Multiple seats in each row depending upon the Ticket type. Row 1 to 3 is for gold category and have 5 seats, row 4 to 7 is for silver category having 7 seats and row 8 to 10 is for platinum category having 12 rows. Price for gold category is Rs.1500, Silver is 1000, and platinum is 500. Based upon this do the following tasks using jagged arrays.



Task-1

- The first entry is the Number of rows.
- For each row, No. of seats will be specified.

Task-2

- Assign element to each seat according to the ticket type.
- If any seat is empty or not booked by the customer, then assign 0.

Lab Task-2

Task-1

Using array and a pointer perform the following for static Jagged Array

- Declare 1-D arrays with the number of rows you will need
- The size of each array (array for the elements in the row) will be the number of columns (or elements) in the row
- Declare a 1-D array of pointers that will hold the addresses of the arrows
- The size of the 1-D array is the number of rows you want in the jagged array

Task-2

Using an array of pointer perform the following for Dynamic Jagged Array

- Declare an array of pointers (jagged array)
- The size of this array will be the number of rows required in the jagged array

- For each pointer in the array allocate memory for the number of elements you want in this row.

Task-3

By using dynamic allocation create a jagged array which will create the following output

Input stream:

```
Enter elements:
1
1 2
1 2 3
```

Output stream:

```
Jagged Array:
1
1 2
1 2 3
```

Lab Task-3

At the beginning of the semester, the students can enroll in a semester by registering process. Then only the passed out students can enroll/register in the next semester. From all the enrolled students in the semester appeared in the exams and got Grade based upon their marks.by considering this scenario perform the below tasks.

Task-1

Receive data on course numbers and grades by semester

Task-2

Compute GPA of students

Task-3

Search for course grades of students

Task-4

Make changes in the records of students

Idea:

- Use two jagged arrays of an identical shape, one for course numbers and the other for letter grades

- The number of rows is the number of semesters
- For each row the number of data elements is the number of courses
- For each user enter number and grade for all the courses

Sample Data

courses

0	“CSC120”	“ENG105”	“PHL110”
1	“BIL101”	“CHM101”	
2	“HIS180”	“PHL181”	“CSC220”

grades

“A+”	“A”	“B+”
“B”	“A”	
“A-”	“B”	“W”

User Input

CSC120 A+ ENG105 A PHL110 B+
BIL101 B CHM101 A
HIS180 A- PHL181 B CSC220 W